From the charged force to the mean memory kernel
A fixed-time bridge for the temporal singlet

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Abstract

The projected memory of the complex SYK temporal singlet has a controlled scalar limit. After energy centering, its annealed trace, multiplied by the interaction order p, converges to $4J_s^2$ times the existing q-Mehler probe kernel with probe parameter q. A marked-site contraction argument retains the dependence of the charged force on the same Hamiltonian. Two finite-system bounds connect this force to orthogonal propagation. Integration gives the first $1/p$ correction to the elementary fermion propagator, recovering the known $\log\cosh$ form in the subsequent small-λ limit. Exact renewal coefficients provide an interface to the operator fluctuations that remain to be calculated.

1 Conventions and the result

Use $\{c_i, c_j^\dagger\} = \delta_{ij}$, $\hbar = 1$, string Hamiltonian $H = H_s$, and the full infinite-temperature, zero-chemical-potential trace $\mathrm{tr}_0 = 2^{-N}\,\mathrm{Tr}$. The neutral Hamiltonian is $H = \sum_{I,J} J_{IJ} C_I^\dagger C_J$, with increasing p-subsets, $C_I = (C_I^\dagger)^\dagger$, Hermitian Gaussian J and covariance

$$\mathbb{E} J_{IJ} J_{KL} = w \delta_{IL} \delta_{JK}, \qquad w = 4^p \frac{J_s^2 N}{p^2} \left(\frac{p!}{N^p} \right)^2 . \tag{1}$$

Overlapping I, J are included. Take $N, p \to \infty$ first, with $p^2/N \to \lambda > 0$; set $q = e^{-4\lambda}$ and $\sigma^2 = J_s^2/\lambda$. Time stays in a fixed compact interval. The charge-block memory below has the scalar limit

$$p\, \mathbb{E}\, \mathrm{tr}_0\, \widehat{\mathcal{M}}(s) \longrightarrow 4 J_s^2 G_q(s), \qquad \widehat{\mathcal{M}}_m(s) = e^{-ihs/2} \mathcal{M}_m(s) e^{-ihs/2}. \tag{2}$$

Here $G_q(0) = 1$; its two-energy measure is specified on page 5. Equation (2) concerns a mean, not an operator limit.

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2 Exact first-return coefficients

For $m > 0$, let $h = H_m$, $K = 1_N \otimes H_{m-1}$, and

$$V\psi = m^{-1/2} \sum_i |i\rangle c_i \psi, \quad P = VV^\dagger, \quad Q = 1 - P, \quad \mathcal{R} = QKV, \quad K_\perp = QKQ. \tag{3}$$

Then $V^\dagger V = 1$ and $h_{\text{eff}} = V^\dagger K V = (1 - r)h$, where $r = p/m$. The preceding bilocal construction gives

$$F(s) = V^\dagger e^{iKs} V, \qquad F'(s) = i h_{\text{eff}} F(s) - \int_0^s \mathcal{M}(s - u) F(u)\, du, \tag{4}$$

$$\mathcal{M}(s) = \mathcal{R}^\dagger e^{iK_\perp s} \mathcal{R}, \qquad G = \mathcal{M}(0) = \frac{A_m}{m} - r^2 h^2 \geq 0, \tag{5}$$

$$A = \sum_i [H, c_i]^\dagger [H, c_i].$$

Define $B_n = V^\dagger K^n V$, $B_0 = 1$, and the ordered recursion

$$b_1 = B_1, \qquad b_n = B_n - \sum_{j=1}^{n-1} b_j B_{n-j}. \tag{6}$$

Insert $1 = P + Q$ between consecutive factors of K and split at the first return to P. This proves the recursion and

$$b_n = V^\dagger K (QK)^{n-1} V, \qquad b_{j+2} = \mathcal{R}^\dagger K_\perp^j \mathcal{R} \quad (j \geq 0). \tag{7}$$

This is the standard noncommuting first-return construction [1], applied to powers of this self-adjoint generator; it avoids building the full orthogonal space.

For an explicit bridge to the derivative singlets, write $M_3 = \sum_i c_i^\dagger \operatorname{ad}_H^3(c_i)$, $T_3 = (M_3 + M_3^\dagger)/2$ and $\mathcal{J}_h X = (hX + Xh)/2$. Direct expansion gives

$$B_2 = A_m/m + (1 - 2r)h^2, \tag{8}$$

$$B_3 = T_{3,m}/m + 3\mathcal{J}_h A_m/m + (1 - 3r)h^3, \tag{9}$$

$$b_3 = T_{3,m}/m + (1 + 2r)\mathcal{J}_h A_m/m - r^2(1 + r)h^3. \tag{10}$$

Consequently $\widehat{\mathcal{M}}(s) = G + isZ_1 + O(s^2)$, with

$$mZ_1 = m(b_3 - \mathcal{J}_h G) = T_{3,m} + 2r\mathcal{J}_h A_m - mr^3 h^3. \tag{11}$$

These are finite-N operator identities. They do not identify the memory parameter s with an external Heisenberg centre time.

3 A same-Hamiltonian force contraction

Let $D_i = [H, c_i]$, $D_i(s) = e^{iHs} D_i e^{-iHs}$ and

$$\mathcal{F}_N(s) = \frac{2p}{N} \sum_i \mathbb{E} \operatorname{tr}_0 D_i^\dagger D_i(s). \tag{12}$$

Its two-energy measure is positive: in an energy basis its weights are $(2p/N)2^{-N} \sum_i \mathbb{E}|(D_i)_{ba}|^2$ at (E_a, E_b). We claim convergence of every fixed polynomial moment to $4J_s^2 \mu_q$, where μ_q is the normalized probe measure below.

Paired force endpoints. Expand $\sum_i \mathbb{E} \operatorname{tr}_0 D_i^\dagger H^u D_i H^v$ by Gaussian Wick contraction, keeping the site i in the two commutators. In graphs where the force endpoints pair together, the corresponding oriented p-set has p choices for i. Except for an expected $O(1)$ choices, this site lies in no other set, including the opposite set on its own chord. Isolate that one-site CAR factor. The unmodified Hamiltonian word contains either n_i or $1 - n_i$, of normalized trace $1/2$. The two commutators remove its creation/annihilation pair and leave the identity there. The remaining odd monomials have the same order; commuting the isolated site through the intervening even word adds no sign. Thus the force trace is exactly twice the original trace.

Deleting one tagged site changes a set size from p to $p - 1$. For any fixed word length the pair-overlap means change by $O(p/N)$, and the probability of a triple overlap is $O(p^3/N^2)$. The remaining chord therefore has the Hamiltonian crossing weight q, including all of its crossings with the exterior chords. It is constrained to join the two force endpoints.

Separately paired force endpoints. If the two endpoints pair with different exterior insertions, the same derivative site must belong to the relevant sets of two independent chords. There are only $O(p^2/N) = O(1)$ such sites on average, instead of p. These diagrams vanish after normalization. For a fixed number of sets, hypergeometric factorial-moment bounds give uniformly bounded exponential moments of all pair overlaps. They control the extra local trace factors 2^{overlap}. With one specified coincidence the same bounds leave an $O(1)$ weighted count; without it they leave $p + O(1)$. Thus exceptional tags and different-chord pairings are $o(p)$ even after local-trace weighting. This also supplies uniform bounds for every fixed higher energy moment. The overlap method is the one underlying the complex chord rules [2]; no independent randomness for D_i is assumed.

The surviving factor is $2p\sigma^2$ times the unit-normalized matter-chord moments. Multiplication by $2p/N$ gives $4\lambda\sigma^2 = 4J_s^2$. Positivity, higher even moment bounds and compact support of the candidate μ_q imply weak convergence by moment determinacy and uniform integrability. The bounded second frequency moment gives equicontinuity, hence convergence of $\mathcal{F}_N(s)$ uniformly on compact time intervals.

4 Removing the orthogonal propagation

On a fixed block put $D = KV - Vh = \mathcal{R} - rVh$, $g = \operatorname{tr}_m G$, $c = \operatorname{tr}_m h^2$. Two useful deterministic bounds are

$$\left|\operatorname{tr}_m e^{-ihs}\mathcal{R}^\dagger(e^{iK_\perp s} - e^{iKs})\mathcal{R}\right| \le \tfrac{1}{2}s^2 \operatorname{tr}_m G^2, \tag{13}$$

$$\left|\operatorname{tr}_m e^{-ihs}(\mathcal{R}^\dagger e^{iKs}\mathcal{R} - D^\dagger e^{iKs} D)\right| \le 2r\sqrt{gc} + r^2 c. \tag{14}$$

For (13), compare K with the block-diagonal $K_0 = Vh_{\text{eff}}V^\dagger + QKQ$. A Q-to-Q difference requires two off-diagonal transitions. Two Duhamel insertions give a triangle of area $s^2/2$. Including the exterior $\mathcal{R}$ factors, each integrand has four rectangular factors with the singular values of $\mathcal{R}$; the other factors are unitary or contractions. Schatten–4 Hölder therefore bounds the normalized trace by $\operatorname{tr}_m G^2$. Equation (14) follows by expanding $D = \mathcal{R} - rVh$ and using Hilbert–Schmidt Cauchy–Schwarz.

The exact second moments, with normalized block trace, are

$$C_m = \mathbb{E}\operatorname{tr}_m h^2 = w\binom{m}{p}\binom{N-m+p}{p}, \tag{15}$$

$$\mathbb{E}\operatorname{tr}_m G = C_m \frac{p(m-p)(N+1)}{m^2(N-m+1)}. \tag{16}$$

In the central window $|m - N/2| \le N^{2/3}$ these imply $\mathbb{E}g = O(p^{-1})$ and $r = O(p^{-1})$ uniformly. Also $\mathbb{E}\operatorname{tr}_m A_m = C_m p(N+1-p)/(N-m+1) = O(p)$ there. The preceding singlet calculation gives the full-trace estimate $\mathbb{E}\operatorname{tr}_0(A - a(Q_f))^2 = O(1)$, with $a(m) = \mathbb{E}\operatorname{tr}_m A_m$. Since $0 \le G \le A_m/m$, the central-window average obeys $\mathbb{E}\operatorname{tr} G^2 = O(p^{-2})$.

For completeness the charge tails are controlled, not discarded. Writing $T(X) = \sum_i c_i^\dagger X c_i$, one has $A_m \le 2T(H_{m-1}^2) + 2mh^2$ and $T(X)^2 \le mT(X^2)$. Gaussian fourth-moment bounds $\mathbb{E}\operatorname{tr}_m h^4 \le 3C_m^2$ then give the conservative estimate

$$\mathbb{E}\operatorname{tr}_m G^2 \le 24(C_{m-1}^2 + C_m^2). \tag{17}$$

The uniform bound $C_m \le (J_s^2/\lambda_N)e^{2\lambda_N}$ and binomial tail $2e^{-2N^{1/3}}$ absorb all polynomial factors; $m < p$ has $H_m = 0$, and the $m = 0$ memory is defined to vanish.

After multiplication by p, (13) is $O(s^2/p)$ and (14) is $O(p^{-1/2})$. Finally $D\psi = m^{-1/2}\sum_i |i\rangle D_i\psi$. Replacing $1/m$ by $2/N$ in the central window has relative error $O(N^{-1/3})$ and is controlled by the positive force mass. This proves (2) from (12).

5 The existing kernel and a second route to its coefficient

In the standard continuous q-Hermite notation, $(a;q)_\infty = \prod_{n\ge 0}(1-aq^n)$, set

$$E(\theta) = \frac{2\sigma\cos\theta}{\sqrt{1-q}}, \qquad \rho_q(\theta) = \frac{(q;q)_\infty |(e^{2i\theta};q)_\infty|^2}{2\pi}, \tag{18}$$

$$K_r(\theta,\phi) = \frac{(r^2;q)_\infty}{|(re^{i(\theta+\phi)};q)_\infty (re^{i(\theta-\phi)};q)_\infty|^2}, \tag{19}$$

$$d\mu_r = \rho_q(\theta)\rho_q(\phi)K_r(\theta,\phi)\,d\theta\,d\phi, \tag{20}$$

$$G_r(s) = \int_{[0,\pi]^2} e^{is[E(\phi)-E(\theta)]}\,d\mu_r. \tag{21}$$

This is the established probe kernel [2]. The calculation above specifies why the present same-J force selects $r = q$.

There is an independent identity of weighted measures:

$$\lim_{r\uparrow 1} \frac{[E(\theta)-E(\phi)]^2}{1-r}\,d\mu_r = 2\sigma^2\,d\mu_q. \tag{22}$$

To verify it, isolate the $n = 0$ denominator factors and use

$$|1-e^{i(\theta+\phi)}|^2|1-e^{i(\theta-\phi)}|^2 = 4(\cos\theta-\cos\phi)^2. \tag{23}$$

The numerator divided by $1-r$ tends to $2(q;q)_\infty$; the remaining denominator is precisely the one at $r = q$. Since $(q;q)_\infty = (1-q)(q^2;q)_\infty$, the factor in (22) follows. This is an almost-everywhere density limit and a weak measure limit, not equality on the diagonal: the left density is zero there for every $r < 1$. For r near one, $|1-re^{i\alpha}|^2 \ge r|1-e^{i\alpha}|^2$ bounds the cancelled factors, and the remaining q products are bounded away from zero. Dominated convergence rules out a residual diagonal delta mass.

This identity checks the coefficient and kernel independently; it does not justify substituting a size-one fermion into a formula derived for double-scaled independent probes. The tagged-force argument on page 3 establishes the required microscopic limit.

For numerical evaluation use the Jacobi operator $(T_q)_{n-1,n} = \sigma\sqrt{(1-q^n)/(1-q)}$. Then

$$G_q(s) = \sum_{n\ge 0} q^n |\langle n|e^{-isT_q}|0\rangle|^2. \tag{24}$$

In the subsequent small-λ limit, the known probe contraction with $\Delta = 1$ and energy scale $\sigma\sqrt{-\log q} = 2J_s$ gives $G_q(s) \to \operatorname{sech}^2(2J_s s)$.

6 The first correction to elementary propagation

Define the averaged propagator with unit equal-time normalization,

$$f_N(t) = \frac{2}{N} \sum_i \mathbb{E}\,\mathrm{tr}_0\, c_i^\dagger c_i(t), \qquad f_N(0) = 1. \tag{25}$$

The commutator ad_H is self-adjoint in the trace Hilbert space. Twice differentiating the finite-system expression therefore gives the exact identity

$$-p f_N''(t) = \mathcal{F}_N(t). \tag{26}$$

Gaussian sign symmetry gives $\mathbb{E} f_N'(0) = 0$ and an even real averaged correlator. Compact-time convergence of the force can now be integrated, with no exchange of an infinite Taylor series:

$$p[1 - f_N(t)] \longrightarrow 4J_s^2 \int_0^{|t|} (|t| - s) G_q(s)\, ds. \tag{27}$$

Taking $\lambda \downarrow 0$ afterwards yields

$$\lim_{\lambda \downarrow 0} \lim_{\mathrm{DS}} p[1 - f_N(t)] = \log\cosh(2J_s t). \tag{28}$$

This recovers the established large-interaction-order propagator structure [3]. In that comparison the interaction order is $q_{\mathrm{MS}} = 2p$, whereas our $q = e^{-4\lambda}$ is the chord deformation parameter. The energy scale in the hyperbolic function is $2J_s$. The result is a fixed-time first correction; exponentiating it does not establish a resummed formula on times of order p.

7 Checks and finite-λ values

Three exact rational-complex CAR systems verify the ordered renewal recursion through order six and both first-memory-moment formulas. Three additional systems check (13)–(14) at four times each. An independent isolated-tag test has 130 words, including 17 nonzero traces. These check the local algebra; the all-order contraction proof is on page 3.

The weighted Mehler identity passes 25 exact chord-polynomial tests with $u + v \le 8$ and nine off-diagonal product checks at 65 digits. For $J_s = 1$, the normalized memory values are:

λ	$s = 0.25$	$s = 0.5$	$s = 1$
0.25	0.8540298823	0.5342098536	0.0949863857
0.10	0.8166877704	0.4634948525	0.0780373626
0.025	0.7943841554	0.4304109293	0.0722466308
0.01	0.7896505803	0.4241100990	0.0712705372
$\lambda \to 0$	0.7864477330	0.4199743416	0.0706508249

8 The next operator problem

The scalar mean has now been separated from the fluctuations that can carry collective singlet information. At zero memory time the known conserved subtraction gives

$$mG - \left[a_m + c_m(h^2 - C_m) - mr^2h^2\right] = Y_m, \tag{29}$$

where $Y = A - a(Q_f) - c(Q_f)(H^2 - C(Q_f))$ is the previously computed dynamic quadratic singlet. At first order, Eq. (11) supplies the exact combination of the cubic singlet, an energy descendant of A, and a conserved cubic polynomial. Thus the memory expansion starts from already identified microscopic fields.

The next calculation is a covariance of the centered memory with two independent memory arguments and an external centre time, after specifying the conserved projection and normalization. One should first obtain its first-return moment coefficients from (6), retain the same-J contractions between the two insertions, and then establish spectral bounds for finite memory arguments. Replacing either insertion by its annealed mean would erase precisely this collective information.

This supplies a concrete route towards the temporal-bilocal to bulk meson interface. The present mean kernel alone fixes neither a spatial translation generator nor a meson mass spectrum. The memory argument, external centre time and bulk internal light-front variable remain distinct. The full de Sitter holographic construction is still open.

Reproducibility. The stage directory stores the two calculation records, versions, seeds and input SHA-256 values. Jacobi calculations use a vacuum Taylor truncation bound below 10^{-13} and a size increase of 12; for $\lambda = 0.25, 0.1$ they are also checked against independent 128/192-node Mehler quadrature. Refinement assesses floating-point and quadrature error, not interval certification. The table retains finite-λ deviations from the limiting function.

Two initial failures are preserved: an integer conversion lost half-integral coefficients in an exact-matrix helper, and random tagged words did not meet the nonzero-coverage target. The helper now preserves exact dyadic rationals, and ten directed nonzero words were added without lowering the target. No failed output was replaced by a passing claim. This stage is saved locally, without Git commit, push or a release archive.

References

[1] J. Bourgain, F. A. Grünbaum, L. Velázquez and J. Wilkening, *Quantum recurrence of a subspace and operator-valued Schur functions*, arXiv:1302.7286, pp. 6–7.

[2] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev–Ye–Kitaev model in the double scaling limit*, arXiv:2006.13983v2, Secs. 4 and 7.

[3] J. Maldacena and D. Stanford, *Comments on the Sachdev–Ye–Kitaev model*, arXiv:1604.07818, Sec. 2.4 and Eq. (3.161).